

MAT 017B Quiz 3

August 14, 2018

Name: Key

SID: _____

1. Evaluate the following indefinite integrals.

a. $\int 3x \sin x \, dx$

$$u = 3x \quad dv = \sin x \, dx$$

$$du = 3 \, dx \quad v = -\cos x$$

$$\int 3x \sin x \, dx = -3x \cos x + \int 3 \cos x \, dx$$

$$= \boxed{-3x \cos x + 3 \sin x + C}$$

b. $\int x^2 \ln x^2 \, dx$

① $\int x^2 \ln x^2 \, dx = \int 2x^2 \ln x \, dx$

$$u = \ln x \quad dv = 2x^2 \, dx$$

$$du = \frac{1}{x} \, dx \quad v = \frac{2}{3} x^3$$

$$= \frac{2}{3} x^3 \ln x - \int \frac{2}{3} x^2 \, dx$$

$$= \boxed{\frac{2}{3} x^3 \ln x - \frac{2}{9} x^3 + C}$$

② $u = \ln x^2 \quad dv = x^2 \, dx$

$$du = \frac{2}{x} \, dx \quad v = \frac{1}{3} x^3$$

$$= \frac{1}{3} x^3 \ln x^2 - \int \frac{2}{3} x^2 \, dx$$

$$= \boxed{\frac{1}{3} x^3 \ln x^2 - \frac{2}{9} x^3 + C}$$

③ $w = x^2 \Rightarrow x = \sqrt{w}$

$$dw = 2x \, dx$$

$$\int x^2 \ln x^2 \, dx = \int \frac{1}{2} \sqrt{w} \ln w \, dw$$

$$u = \ln w \quad dv = \frac{1}{2} \sqrt{w} \, dw$$

$$du = \frac{1}{w} \, dw \quad v = \frac{1}{3} w^{3/2}$$

$$= \frac{1}{3} w^{3/2} \ln w - \int \frac{1}{3} w^{1/2} \, dw$$

$$= \boxed{\frac{1}{3} x^3 \ln x^2 - \frac{2}{9} x^3 + C}$$

c. $\int \frac{3x+3}{(x-1)(2x+1)} \, dx$

$$\frac{3x+3}{(x-1)(2x+1)} = \frac{A}{x-1} + \frac{B}{2x+1}$$

$$3x+3 = A(2x+1) + B(x-1)$$

$$3x+3 = 2Ax + A + Bx - B$$

$$2A + B = 3 \quad A = 2$$

$$A - B = 3 \quad B = -1$$

$$\underline{\quad\quad\quad} 3A = 6$$

$$= \int \frac{2}{x-1} \, dx - \int \frac{1}{2x+1} \, dx$$

$$= \boxed{2 \ln|x-1| - \frac{1}{2} \ln|2x+1| + C}$$